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A semantic bridge from questions to answers via Chu duality in a fibration attention mechanism

Gary Nan Tie, Nov 2, 2025

Abstract

In a LLM database retrieval paradigm, Q&A seeks a semantic handshake via intermediary keys. A key-value pairing $p: K \rightarrow V$ gives a value v in V semantics via its inverse image under p , like looking up the definition of a word in a dictionary. Keys can also be used to give meaning to queries via a neural network $d: K \rightarrow Q$. Note that semantics p and d are neural networks to be learned. Moreover, we can learn a fibration lifting τ that connects p and d in a semantic handshake, facilitating meaningful answers to queries. Technically, d and p are Chu adjoints under τ , and the value returned to a query q is approximately $p(c(q)) \in V$, where $c: Q \rightarrow K$ is a section of d .

Introduction

In asking a question, presumably there is intent in requesting information. This intent is communicating meaning not random gibberish. Likewise an answer in reply is expected to correspond somewhat in semantic context, a semantic handshake is desired so to speak. The transformer attention mechanism based on a query-key-value database retrieval paradigm introduces keys as intermediaries between queries and values; giving meaning to values. We seek keys that are similar to a given query.

Keys can give semantics to queries as well as values. Recall a key-value pairing $p: K \rightarrow V$ gives a value $v \in V$ key-semantics via its inverse image under p ($\{k \in K \mid p(k) = v\}$), like looking up the definition of a word in a dictionary. Keys can also give meaning to queries via a neural network $d: K \rightarrow Q$. Note that semantics p and d are neural networks to be learned. Moreover, we can learn a fibration lifting τ that connects p and d in a semantic handshake, facilitating answers to queries. Technically, d and p are Chu adjoints under τ . Fibration liftings determine sections c and s of semantics d and p respectively. Given query q , answer value v , where $c(q)$ and $s(v)$ are close keys.

Query updating in transformer attention involves a linear transformation of values followed by an unconstrained nonlinear feed forward network. In contrast, Chu-former applies structured fibration lifting networks instead, creating sections to semantics that bridge queries and values.

Transformer attention via Chu duality

Data: queries $q_i \in Q \subseteq \mathbb{R}^d$, $i \in [n]$

key-value pairs $(k_j, v_j) \in K \times V \subseteq \mathbb{R}^d \times \mathbb{R}^d$, $j \in [m]$

Let score be a similarity function between keys and queries,
eg scaled dot-product or more generally a reproducing kernel.

Fix $j \in [m]$, let $\alpha_{ij} = \text{softmax}_{i \in [n]} \text{score}(k_j, q_i)$

then $0 \leq \alpha_{ij} \leq 1$ and $\sum_{i=1}^n \alpha_{ij} = 1$.

Learn NN $p: K \rightarrow V$ such that $p(k_j) \approx_\epsilon v_j$, $j \in [m]$.

Learn NN $d: K \rightarrow Q$ such that $d(k_j) \approx_\epsilon \sum_{i=1}^n \alpha_{ij} q_i$, $j \in [m]$.

Fix $i \in [n]$, let $\beta_{ij} = \text{softmax}_{j \in [m]} \text{score}(k_j, q_i)$

then $0 \leq \beta_{ij} \leq 1$ and $\sum_{j=1}^m \beta_{ij} = 1$.

Let $j^* = \arg \max_j \beta_{ij}$

$\{\beta_{ij}\}_{j=1}^m$ is a distribution over $\{k_j\}_{j=1}^m$.

$$\begin{array}{ccc}
 K \otimes K & \xrightarrow{p \otimes 1} & V \otimes K \\
 1 \otimes d \downarrow \quad \tau \nearrow & & \downarrow 1 \otimes d \\
 K \otimes Q & \xrightarrow{p \otimes 1} & V \otimes Q
 \end{array} \quad (†)$$

Given $d: K \rightarrow Q$ and $p: K \rightarrow V$, learn NN [1]

$\tau: K \otimes Q \rightarrow V \otimes K$ such that $(†)$ commutes up to $\approx_\varepsilon$ i.e.

$$i) \quad \tau \begin{bmatrix} k_j \\ d(k_{j'}) \end{bmatrix} \approx_\varepsilon \begin{bmatrix} p(k_j) \\ k_{j'} \end{bmatrix} \quad j, j' \in [m]$$

$$ii) \quad \pi_1 \tau \begin{bmatrix} k_j \\ q_i \end{bmatrix} \approx_\varepsilon p(k_j) \quad j \in [m], i \in [n]$$

$$iii) \quad d \pi_2 \tau \begin{bmatrix} k_j \\ q_i \end{bmatrix} \approx_\varepsilon q_i \quad j \in [m], i \in [n]$$

Learn $c: Q \rightarrow K$ such that $c(q_i) \approx_\varepsilon \pi_2 \tau \begin{bmatrix} k_{j^*} \\ q_i \end{bmatrix} \in K$

By iii) $d(c(q_i)) \approx_\varepsilon q_i$, so on $\{q_1, \dots, q_n\}$

c is a section of d .

[i) is a Chu adjointness condition between d and p .]
See [3].

Similarly,

$$\begin{array}{ccc}
 K \otimes K & \xrightarrow{d \otimes 1} & Q \otimes K \\
 \downarrow \text{1} \otimes p & \searrow \sigma & \downarrow \text{1} \otimes p \\
 K \otimes V & \xrightarrow{d \otimes 1} & Q \otimes V
 \end{array}
 \quad \left(\begin{array}{c} \neq \\ \downarrow \end{array} \right)$$

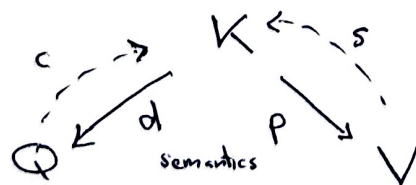
$$i) \quad \sigma \begin{bmatrix} k_j \\ p(k_{j'}) \end{bmatrix} =_{\varepsilon} \begin{bmatrix} d(k_j) \\ k_{j'} \end{bmatrix} \quad j, j' \in [m]$$

$$ii) \quad \pi_1 \sigma \begin{bmatrix} k_j \\ v_{j'} \end{bmatrix} =_{\varepsilon} d(k_j) \quad j, j' \in [m]$$

$$iii) \quad p \pi_2 \sigma \begin{bmatrix} k_j \\ v_{j'} \end{bmatrix} =_{\varepsilon} v_{j'} \quad j \neq j' \in [m]$$

$$s(v_j) \triangleq \pi_2 \sigma \begin{bmatrix} k_j \\ v_j \end{bmatrix} \text{ implies } p(s(v_j)) =_{\varepsilon} v_j$$

defines a section $s: V \rightarrow K$ of $p: K \rightarrow V$ on $\{v_1, \dots, v_m\}$.



c is a section of d and s is a section of p

A key $k \in S^{-1}(v_j)$ explains value v_j , in that $s(k) = v_j$,

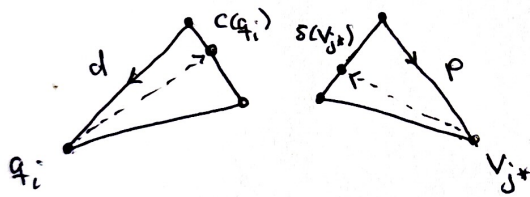
likewise a key $k \in d^{-1}(q_i)$ explains query q_i , in that $d(k) = q_i$.

Keys provide semantics for values and queries.

Let $m: K \times K \rightarrow \mathbb{R}^+$ be a metric on keys in the latent space.

Let $s(v_{j*})$ be the key closest to $c(q_i)$.

In reply to query q_i , answer value v_{j*} .



$$c(q_i) \approx_m s(v_{j*})$$

Semantic Handshake

Query updating:

Recall Rényi α -entropy.

For $0 < \alpha < \infty$, $\alpha \neq 1$

$$H_\alpha(\{p_i\}_{i=1}^n) \triangleq \frac{1}{1-\alpha} \ln \left(\sum_{i=1}^n p_i^\alpha \right)$$

where $0 \leq p_i \leq 1$ and $\sum_{i=1}^n p_i = 1$.

For $\alpha = 0, 1, \infty$, $H_\alpha \triangleq \lim_{\gamma \rightarrow \alpha} H_\gamma$

Choose hyperparameter $\alpha \geq 0$.

Define $E_T \triangleq \frac{1}{n} \sum_{i=1}^n H_\alpha^i$,

where $H_\alpha^i \triangleq H_\alpha(\{\beta_{ij}\}_{j=1}^m)$.

Let query update $q'_i \triangleq \sum_{j=1}^m \beta_{ij} v_j$ and T' be the

corresponding lifting using the q'_i (so $d'(k_j) = \sum_{i=1}^n \alpha'_{ij} q'_i$

where $\alpha'_{ij} = \underset{i \in [n]}{\text{softmax score}}(k_j, q'_i)$), likewise

corresponding lifting $\sigma'(d')$.

Note that E_{τ} depends on $\text{score}(k_j, q_i)$

and $E_{\tau'}$ depends on $\text{score}(k_j, q'_i)$.

If $E_{\tau'} < E_{\tau}$

assign $q_i := q'_i$, $\tau := \tau'$ and $\sigma := \sigma'$ and repeat...

Stop updating when $E_{\tau'} \geq E_{\tau}$ or

$E_{\tau'}$ is sufficiently small.

Each iteration we first update the current query, then

learn the corresponding fibration liftings to create sections

to semantics, continuing until entropy is sufficiently small,

at which point we output value v_{j*} in answer to query q_i .

→

Summary

Keys provide semantics for values and queries,
given by NNs $p: K \rightarrow V$ and $d: K \rightarrow Q$.

p and d are Chu duals with respect to

fibration liftings τ and σ that provide

sections $c: Q \rightarrow K$ and $s: V \rightarrow K$ for

d and p respectively. c and s create a

semantic bridge from queries to values.

Query updating is achieved by entropy reduction

of fibration lifting τ .



References

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Transformer attention uses statistical correlation to update queries, and fibration abduction attention on top, takes into account key-value semantics, finding a key to explain the value returned. The latter has a quantum interpretation, indeed a fibration lifting is an entanglement. Entanglements (inner loop) can be refined using a Renyi α -entropy criterion, as part of a simultaneous query and key update (outer loop). Thus the value returned then reflects refinement of both query-key similarity and key-value semantics.

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